Storage Services

Use Case 3: Backup and Restore

Scenario: Implement a backup strategy for critical data.

Tasks:

o Use AWS Backup to back up RDS and S3 data.

o Restore data to a new region as part of disaster recovery testing.

o Analyze costs associated with the backup.

Step 1: Set Up AWS Backup for RDS and S3

1.1. Enable AWS Backup

- Log in to your **AWS Management Console**.
- Navigate to **AWS Backup**.
- In the AWS Backup Dashboard, click on **Settings**.

- Enable backup services for **RDS** and **S3** by selecting them under **Backup vault access permissions**.

1.2. Create a Backup Plan

- Go to **Backup Plans** in the AWS Backup console.
- Click **Create Backup Plan** and choose:
  - **Build a new plan** or use an **existing template**.
- Define the backup schedule:
  - Select a backup frequency (e.g., daily or hourly).
  - Specify the retention period for your backups (e.g., 30 days).
- Choose a **Backup Vault** (a secure container to store backups). If none exists, create one:
  - Name the vault.
  - Enable encryption (using AWS KMS).

- Save the backup plan.

1.3. Assign Resources to the Backup Plan

- In the AWS Backup console, go to **Resource Assignments**.
- Click **Assign Resources** and:
    - Provide a name for the assignment.
    - Select **RDS** and **S3** as the resource types.
    - Choose specific RDS databases and S3 buckets.
- Review and save the assignment.

Step 2: Test Backup and Restore

2.1. Verify Backup Creation

- Go to **Backup Vaults**.
- Confirm that backups for your RDS databases and S3 buckets are successfully created as per the schedule.

2.2. Restore Data in a New Region

- Go to **Backup Jobs** in the AWS Backup console.
- Select a completed backup job for RDS or S3.
- Click **Restore** and follow these steps:
  - Choose the **new region** where you want to restore the backup.
  - For RDS:
    - Provide the database instance details for restoration (e.g., instance class, VPC, security group).
  - For S3:
    - Specify the target bucket or create a new bucket in the new region.
- Confirm and start the restore job.

2.3. Validate the Restoration

- For RDS: Check the restored database instance in the target region and validate the data.
- For S3: Access the restored bucket and verify the files.

Step 3: Analyze Backup Costs

3.1. Cost Breakdown

- Navigate to **AWS Cost Explorer** in the AWS Management Console.
- Filter costs by:
  - **Service**: Select AWS Backup, RDS, and S3.
  - **Region**: View costs for both the source and target regions.
  - **Time Period**: Analyze costs over a specific period (e.g., monthly or daily).

3.2. Key Cost Components

- **Storage Costs**:

- Backup vault storage charges (standard vs. cold storage).

- **Restore Costs**:
  - Data transfer charges between regions.
  - Restore jobs (RDS database instance and S3 bucket).
- **Additional Charges**:
  - AWS KMS charges for encryption.
  - Costs for retention beyond the free tier.

3.3. Generate Cost Reports

- Use **Cost and Usage Reports (CUR)** in the Billing Console:
  - Create a report to track AWS Backup usage and costs.
  - Export the report to S3 for analysis using tools like Amazon Athena or AWS QuickSight.

Step 4: Implement Disaster Recovery Testing

4.1. Simulate a Disaster

- Assume primary region resources (RDS and S3) are unavailable.

4.2. Validate Restore Process

- Access the restored data in the new region.
- Test application functionality using the restored RDS database and S3 bucket.

4.3. Document the Process

- Create a Standard Operating Procedure (SOP) for disaster recovery:
  - Include details of the restore process.
  - Add timelines for RTO (Recovery Time Objective) and RPO (Recovery Point Objective).

Step 5: Optimize and Monitor

5.1. Optimize Backup Strategy

- Adjust backup frequency and retention policies based on cost analysis.
- Use lifecycle policies to transition backups to lower-cost storage tiers (e.g., from standard to cold storage).

5.2. Set Up Monitoring

- Enable **AWS CloudWatch** to monitor backup jobs and storage usage.
- Use **AWS Backup Audit Manager** to track compliance with backup policies.